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B. Atomizing Air

Air Compressor Pressurizes air to be used in the atomization of oil. Not present if shop air (plant air) is to be used. Typical air pressures range up to 100 psi.

Air Regulator Controls fluctuations in the air pressure from shop air (plant air) line. Not present if air compressor is used.

Air Tank Dampens pulsations from air compressor.

Air Switch Permissive- Low air pressure disables electrical control circuit.

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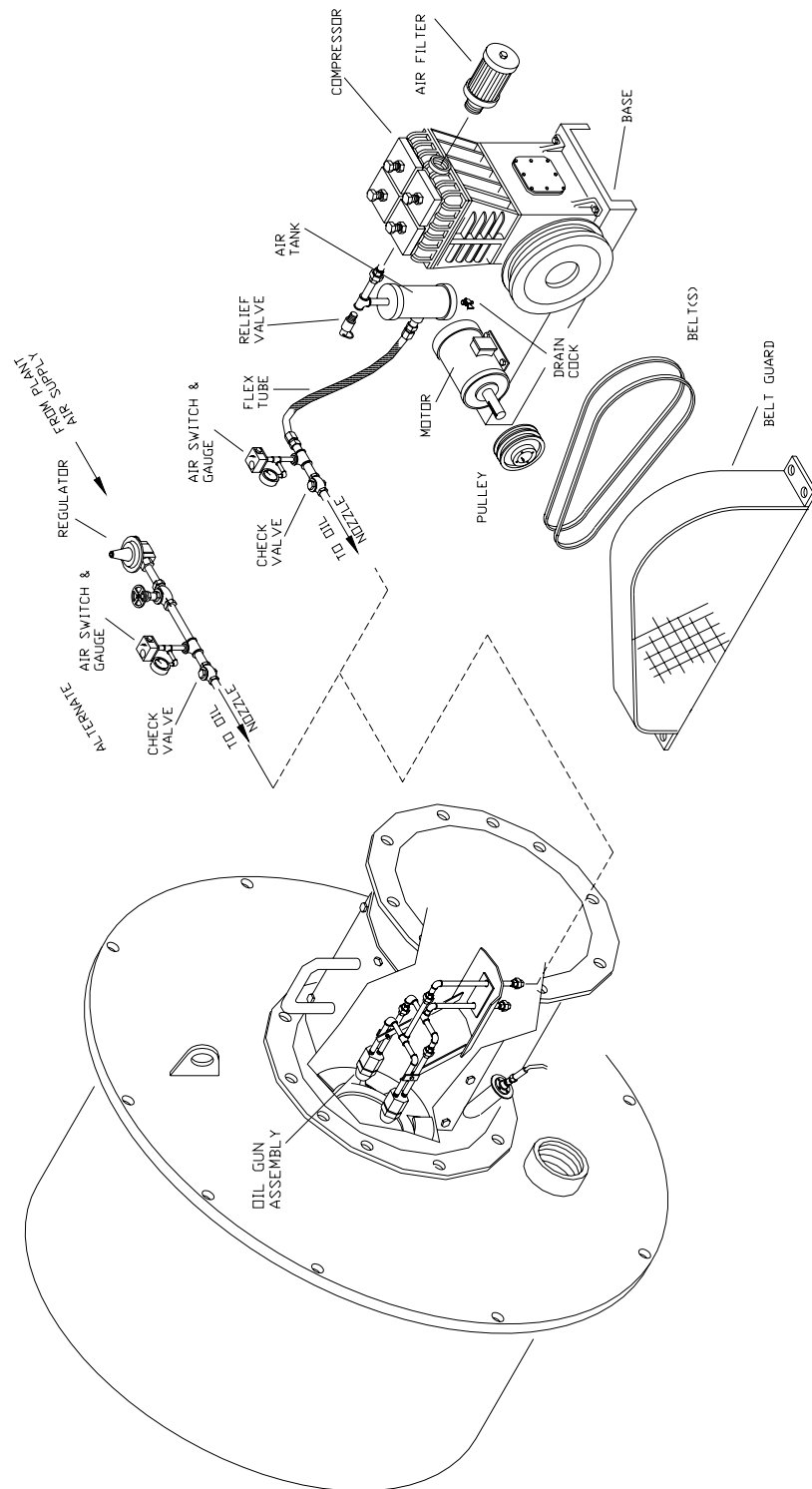


Fig. 7.2

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C. Light Oil

Inlet Strainer	Filters out any foreign matter in the fuel oil supply.
Oil Pump	Increases the pressure of the fuel oil supply to the required pressure. See oil pump manual for exact inlet pressure ratings. Caution: Some pumps will leak through shaft seal if rated inlet pressure is exceeded.
Oil Regulator	A back supply pressure regulator that maintains constant oil pressure to the flow control valve. Excess oil is recirculated back to a tank or the pump inlet. The set pressure is a function of firing rate and required nozzle pressure. Typical supply pressures to nozzles are 40-80 psig for Delevan nozzles and 50-100 psig for Monarch nozzles. (Not always supplied, may be integral to pump).
Low Oil Pressure Switch	Opens under a low oil supply pressure condition. Will not allow the burner to operate under this condition.
Safety Shutoff Valves	Open to admit oil to the main burner, but only after a pilot has been proven. Electrically actuated. If the flame signal drops off, the oil valve will shut to insure proper, safe combustion (May be equipped with a proof of closure switch depending on insurance/code requirements).
Flow Control Valve	Throttles the amount of regulated fuel oil supply to the main burner from low to high fire.

D. Heavy Oil Additional equipment provided for heavy oils.

Fuel Oil Preheaters	Electric and/or Steam. Used to heat heavy (No. 4 thru No. 6) fuel oils, tallow, and waste oil. An electric preheater is a thermostatically controlled, low watt density preheater, for cold oil starting. A permissive low oil temperature switch is built in to prevent starting on a low temperature condition. A steam preheater is an indirect steam heater with a capacity to heat the fuel oil from approximately 100°F to 230°F. A steam preheater is used to heat heavy oils once the boiler reaches its normal operating steam pressure. When used in combination fuel burners, the electric heater is used as a trim heater, maintaining a constant supply temperature.
Heat Trace Oil Solenoid Valve (Heavy Oil Only)	Permits the circulation of hot oil in the fuel oil supply system during boiler/burner off periods. This is done to keep the oil in the line hot for good atomization and flow during burner operation.
Temperature Control Valve	Regulates steam flow to heater on the basis of oil temperature entering the heater.
Temperature Switch	Permissive- switch interrupts control circuit on high temperature of oil.
Steam Regulator	Supplied when boiler operating pressure exceeds 150 psig.
Steam Trap	Collects and discharge condensate from steam heater.

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Special Notes:

- Most burners utilize a gas pilot ignition, whether the main burner operates on gas or oil.
- Atomizing air (in some instances, atomizing steam) is used to burn all fuel oils in Johnston burners. Atomizing air can be supplied by an air compressor furnished by Johnston Boiler Company with the burner, or by shop air (plant air) available in the boiler room. If shop air is to be used, an air regulator must be present to maintain constant air pressure.
- Oil pilot (sometimes referred to as a “diesel ignition”) uses No. 2 oil for pilot flame.

Fuel Train

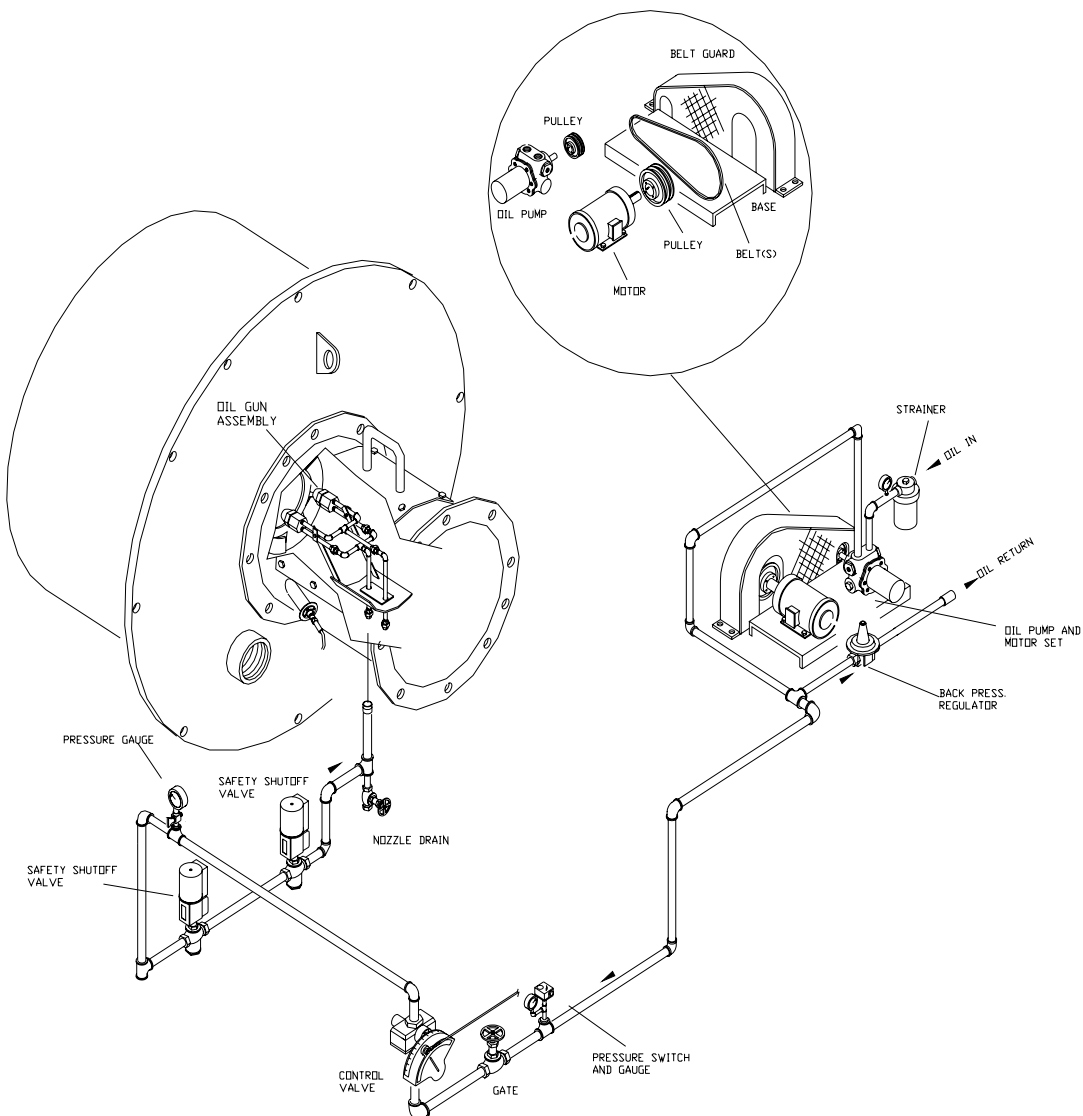


Figure 7.3: Typical #2 Oil Train

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Fuel Train

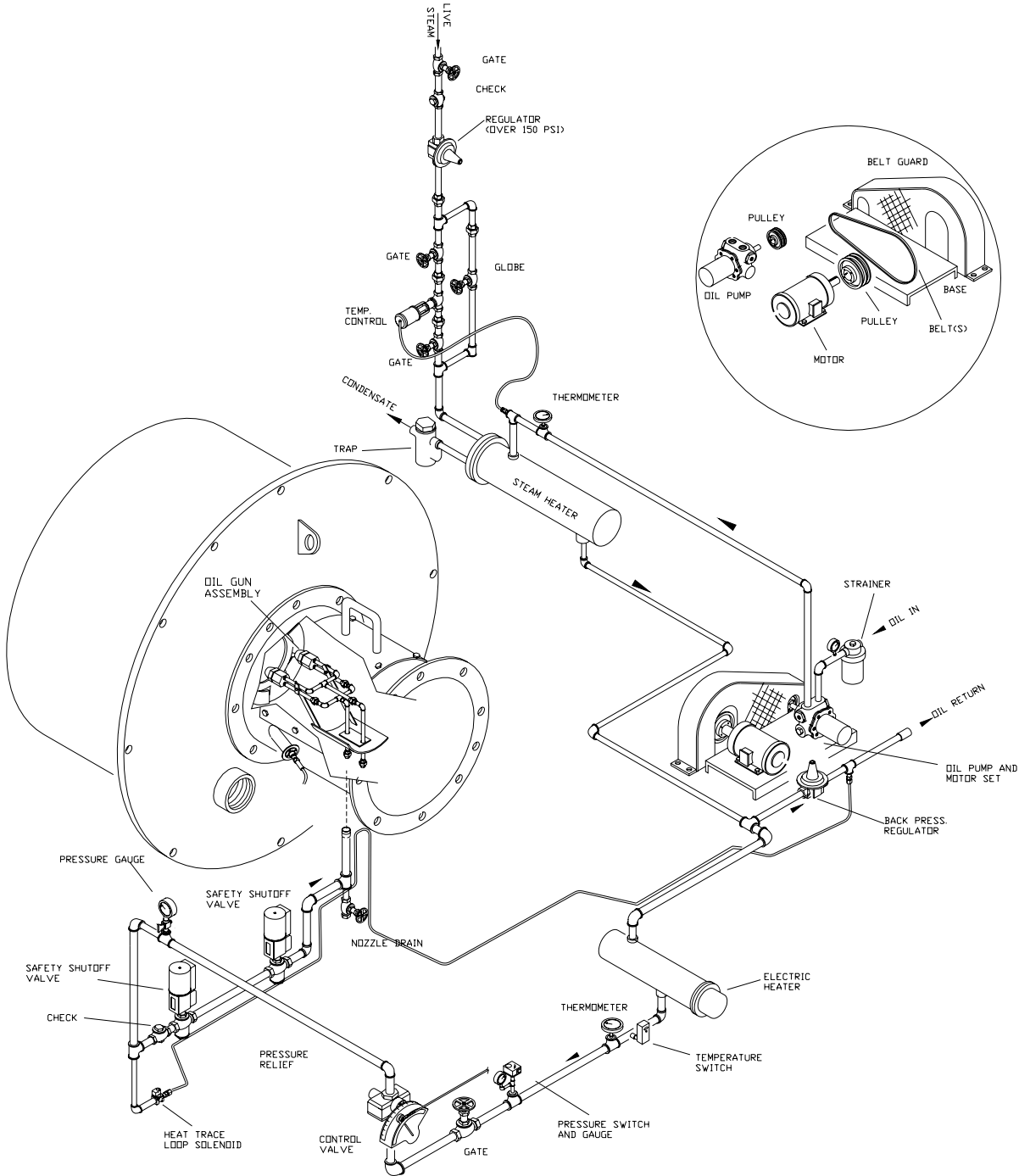


Fig 7.4 Typical Heavy Oil Train

Section 8: Field Installation (If not factory installed)

A. General Information

The following are general recommendations for assembly and installation of a Johnston S style burner. Refer to Section 6 to familiarize yourself with parts and terminology. It is strongly recommended that a qualified field service technician familiar with Johnston products be present to supervise or do the work. These burners will be heavy and unlikely to be lifted by hand. You will need appropriate lifting equipment such as fork lifts, cranes or cherry pickers to position the burner in place. Local Code requirements must also be followed. These legal requirements take precedence over information found in this manual.

B. Installation Recommendations

Your Johnston S style burner will be shipped in multiple pieces and will need to be field assembled at installation. If your burner arrives fully pre-assembled, it will need to be partially broken down prior to installation. The following steps are for reference only and may be altered, skipped or rearranged on the basis of field advice for your application.

- 1) If not done so, break down burner into multiple pieces; **A.** The refractory sleeve; **B.** The plenum and blower transition, blower, inlet housing w/damper and silencer. This requires removal of nuts, bolts and all interconnecting wire, linkage, tubing and pipe.
- 2) Install gasket on refractory sleeve mounting flange, if shipped loose.
- 3) Insert refractory sleeve into furnace opening, plumb, level and attach nuts and washers.
- 4) If excess space between the firing tube and furnace opening exists, fill the gap with a refractory type of insulation blanket, block or mud.
- 5) Attach plenum and transition assemblies via bolts and nuts.
- 6) Attach blower assembly with bolts and nuts.
- 7) Grout and anchor blower assembly to suitable foundation.
- 8) Connect inlet housing, damper and finally the silencer.
- 9) Connect or install all other devices as required for your application and purchase.

- 10) Electrical supply lines should be properly installed to the burner. It is recommended that fuel supply lines be properly valved, so the supply lines to the burner can be shut off when it is not in use or when preventive maintenance or repairs to the burner are being made. A "dirt trap" should be installed in the gas supply line, just prior to entering the main gas train of the burner. A means to disconnect and lockout the power supply to the burner control panel should also be provided. Installation of burner fuel and electrical power supply piping and lines should be checked, so as not to interfere with operation of any component part of the burner, or interfere with the ability to perform preventive maintenance or make repair. Special attention should be given to making sure that no piping or lines interfere with chamber doors, hatches, or removable components.

C. Boiler Room Air Supply

The starting point in any combustion system is the supply of fresh air. To avoid serious problems, the boiler must have an adequate supply of fresh air and a supply system that does not affect the boiler operation. An adequate, proper and consistent supply of fresh air must be supplied into the boiler room for burner combustion air. A rule of thumb is to provide four to six square inches of open, unrestricted area for every boiler horsepower. Example: For a 75 HP boiler, 300 to 450 square inches of open unrestricted area are recommended.

How much air is required?

In general, the following formulas have been developed to determine the amount of combustion air required for any boiler room with a package firetube boiler firing gas or oil fuels:

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$$\text{Combustion Air} = HP \cdot 8 \frac{\text{ft}^3}{\text{min} \cdot HP}$$

$$\text{Ventilation Air} = HP \cdot 2 \frac{\text{ft}^3}{\text{min} \cdot HP}$$

$$\text{Total Air Req} = HP \cdot 10 \frac{\text{ft}^3}{\text{min} \cdot HP}$$

HP refers to the total maximum boiler horsepower in the boiler room.

These calculations are adequate for installations up to 1000 feet above sea level. For any installation above 1000 feet, add 3% air for each 1000 feet.

What size of opening to the outside is required?

The size of the fresh air inlet openings and their location are very important. There should be a minimum of two permanent air supply openings in the walls of the boiler room. Whenever possible, they should be at opposite sides of the boiler room and no higher than 7 feet above the floor. This will promote mixing with the air already in the boiler room, proper cooling of the boilers, and tempering of potentially colder outside air prior to its entering the burner for combustion.

The air inlets should be provided with some type of weather protection, but should never be covered with a fine mesh wire screen. This type of covering results in poor air flow and is subject to clogging by dust, dirt, paper, and other small items.

To determine the net free open area required for an opening, divide the total air required (in cubic feet per minute) by the allowable velocity at the opening, from the table below.

Acceptable Air Velocities in Boiler Room	
0-7 ft above floor	250 feet per minute
Above 7 ft above floor	500 feet per minute

The minimum opening to the outside should be one square foot

Air Ducting

In some applications, the boiler room is located in a building such that it has no outside walls. Many of these applications do not have sufficient excess makeup air in the factory to allow for combustion air requirements. In these cases, there are two possible solutions:

1. Ducting fresh air into the boiler room. Where this is required, the general rules for the size of the wall opening can be used to determine the size of the ducting and wall opening in the boiler room. In addition, the pressure drop through the duct must never exceed 0.05" w.c.
2. Ducting fresh air directly into the boiler. This method of air supply should be avoided whenever possible. The disadvantages of this type of system far exceed any perceived advantages. If used, the ducting becomes a part of the boiler system and can affect the stability of combustion due to varying weather conditions, wind direction and velocity, humidity, and temperature. Changes in temperature can lead to massive CO production, soot formation, and unstable, unsafe combustion.

If direct ducting is to be used, we suggest the following steps be followed:

1. Each boiler has its own, completely separate, fresh air ducting and exhaust stack to minimize combustion problems.
2. Boiler directly connected to fresh outside air ducts must be checked for proper combustion adjustment and operation every three months by a certified package firetube boiler specialist.
3. The duct work supplying the fresh air to the boiler must be sized so that it has a maximum pressure drop of 0.05" w.c. at maximum flow.

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4. The fresh air supply should have an electric, hot water, or steam heater to temper cold outside air to at least 50 deg F.
5. If the application is utilizing a low emission with flue gas recirculation, do not use direct ducted outside air. The potential problems associated with a standard burner are intensified with a low emission burner.

